RACHIT KUMAR TIWARI

Software Engineer | Full-Stack & AI-Integrated Systems

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Professional Summary

Third-year Computer Science undergraduate and full-stack developer who has shipped five production-grade, AI-integrated applications spanning computer vision, financial analytics, and document processing. Proficient in React, Next.js, FastAPI, and modern AI APIs (OpenAI, Gemini, Mistral), with hands-on experience designing secure, containerized systems end-to-end from database to deployment.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C

Frontend: React, Next.js, Electron, HTML5, CSS3, Tailwind CSS, WebAssembly, Web Streams API

Backend & APIs: Node.js, FastAPI, REST APIs, JWT, OAuth, bcrypt, WebSocket

Databases, Cloud & DevOps: MongoDB, Firebase, Supabase (Auth/PostgreSQL/Storage), Vercel, Docker, Git/GitHub

AI & Computer Vision: OpenAI API, Gemini, Mistral AI, RAG, OpenCV, MediaPipe, Real-Time Object Detection

Project Experience

•**MedCheck** *Aug 2026*

AI-Powered Clinical Drug Interaction & Safety Platform React, FastAPI, Python, Mistral AI, Docker

https://github.com/rachts/MEDCHECK | https://medcheck-official.vercel.app

– Architected a FastAPI backend powering a 17-rule pharmacology engine with Mistral AI fallback, delivering multi-tier severity classification and GI risk scoring across 100+ drug interactions.

– Secured APIs with JWT/bcrypt and containerized the stack via multi-stage Docker Compose, cutting environment setup time to under 2 minutes.

•**DocEasy** *Jan 2026*

Document Management & PDF Processing Platform Next.js 16, TypeScript, Supabase, WebAssembly

https://github.com/rachts/DocEasy | https://doceasy-official.vercel.app

– Engineered a hybrid Next.js architecture using WebAssembly and the Web Streams API, achieving O(1) memory complexity for client-side processing of 250MB PDF payloads.

– Implemented Supabase OAuth and PostgreSQL Row-Level Security with end-to-end TypeScript, reducing reported development errors by 30%.

•**Vitamend** *Aug 2024 – Present*

AI-Driven Medicine Donation & Redistribution Platform Next.js 14, TypeScript, MongoDB, Gemini Vision

https://github.com/rachts/vitamend-org | https://vitamend-ngo.vercel.app

– Built a Gemini Vision OCR pipeline to extract medicine details from label images and automatically flag potentially expired products before approval.

– **1st Place** MCKVIE Innovation Symposium | **National Finalist** Cosmohack (Top 10), HACKVSIT (Top 20).

•**VITALS.IO** *Aug 2026 – Present*

SaaS Unit Economics & Financial Health Analyzer Python, FastAPI, Jinja2, Tailwind CSS, Chart.js

https://github.com/rachts/VITALS.IO | https://vitals-io.onrender.com

– Built a FastAPI-based financial analytics platform that ingests raw CSV data and computes **LTV, CAC, ARPA, churn, gross margin, payback period**, and a 0–10 unit-economics health score.

– Engineered fuzzy CSV header matching and interactive **what-if scenario simulation**, with REST APIs for programmatic financial analysis and startup runway forecasting.

Education

•**MCKV Institute of Engineering, West Bengal** *2024 – 2028 (Expected)*

B.Tech in Computer Science & Engineering SGPA (Sem 1–4): 8.64, 7.67, 8.80, 8.70

•**St Aloysius Orphanage and Day School** *2022*

Class XII

•**St Aloysius Orphanage and Day School** *2020*

Class X

Certifications

HackerRank: SQL (Advanced), SQL (Intermediate), Problem Solving (Intermediate), JavaScript (Intermediate), Node.js (Intermediate), REST API (Intermediate), Python (Basic), Java (Basic) , Software Engineer (Problem solving, SQL, and REST API)